Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

HACKTHON PROJECT ABSTRACT SUBMISSION

COURSE: PROJECT EVALUATION (24CS2235F) A.Y. 2026 - 2027

TITLE OF THE PROJECT	ThreeAddrGen: Source-to-Three-Address Code Translator	
S. No.	**University ID**	**Name**
1	2420090129	D. Ruchit
2	2420090065	P Pranay
3	2420030749	P. Poojith
4	2420030020	V. Harshith
Name of the Guide		K S Niraja

Abstract

A compiler is a program that changes code written in a high-level language into a form that a computer can understand and run. For students, it can be difficult to understand how this process works because a compiler has several stages, and these stages are not always visible.

ThreeAddrGen is a web-based learning tool that helps students understand how a compiler works. It takes a simple C-like program as input and shows how the program is changed into Three-Address Code (TAC). The tool displays the main steps of this process, such as identifying tokens, building a syntax tree, converting expressions into postfix form, and generating intermediate code. Students can follow these steps and see how the code changes at each stage.

The project is developed using Python, HTML, CSS, and JavaScript. It can check the input program for syntax and type errors and supports common programming features such as arithmetic expressions, conditions, loops, and logical expressions. It uses Syntax-Directed Translation to guide code generation and backpatching to manage jump instructions used in conditions and loops.

ThreeAddrGen makes compiler design easier to learn by showing the process in a clear and interactive way. It gives students a chance to try simple programs, observe the output, and better understand how a compiler translates source code into intermediate code.

Signature of the Guide HOD